

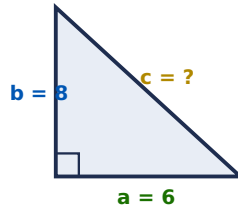
Unit 5 Classwork — Pythagorean Theorem

Hypotenuse · Missing Leg · Converse · Word Problems · Coordinate Distance

A. Find the Hypotenuse

Problem 1

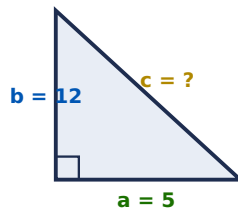
Find the hypotenuse (c) of the right triangle shown.



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Problem 2

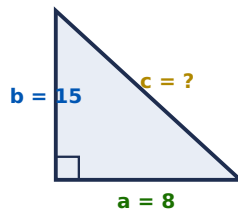
Find the hypotenuse (c) of the right triangle shown.



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Problem 3

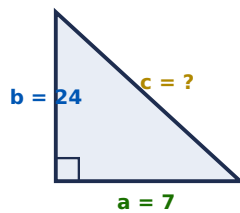
Find the hypotenuse (c) of the right triangle shown.



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Problem 4

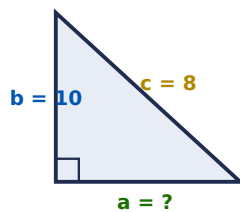
Find the hypotenuse (c) of the right triangle shown.



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B. Find a Missing Leg**Problem 1**

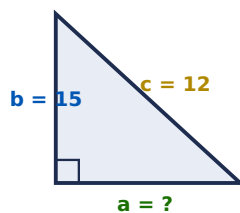
Find the missing leg (a) of the right triangle shown.



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Problem 2

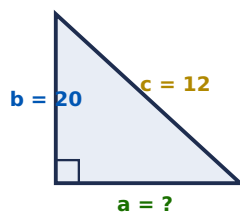
Find the missing leg (a) of the right triangle shown.



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Problem 3

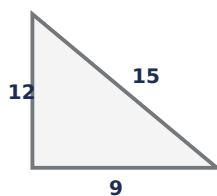
Find the missing leg (a) of the right triangle shown.



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C. Converse — Is It a Right Triangle?**Problem 1**

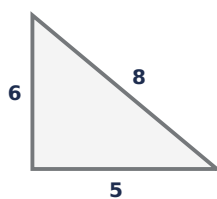
The triangle shown has sides 9, 12, 15. Is it a right triangle?



yes

Problem 2

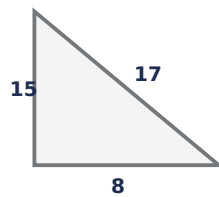
The triangle shown has sides 5, 6, 8. Is it a right triangle?



no

Problem 3

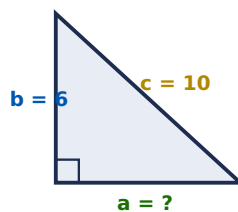
The triangle shown has sides 8, 15, 17. Is it a right triangle?



yes

D. Word Problems**Problem 1**

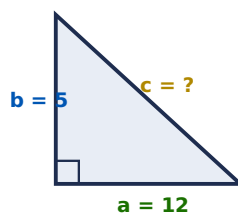
A 10-foot ladder leans against a wall. The base is 6 feet from the wall. How high up the wall does the ladder reach?



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Problem 2

A rectangular garden is 12 m long and 5 m wide. What is the length of the diagonal?

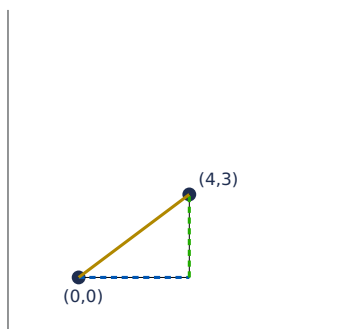


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E. Distance on the Coordinate Plane

Problem 1

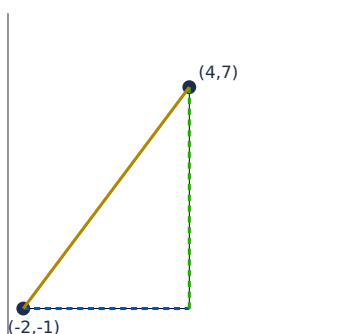
Find the distance between the two points shown.



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Problem 2

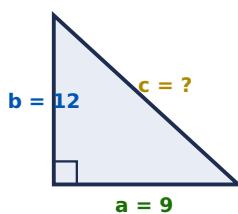
Find the distance between the two points shown.



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F. Mixed Review**Problem 1**

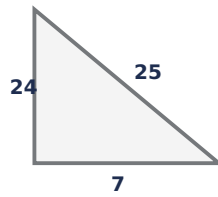
Find the hypotenuse (c) of the right triangle shown.



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Problem 2

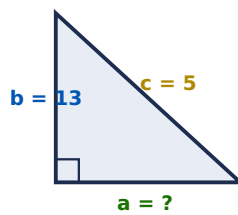
The triangle shown has sides 7, 24, 25. Is it a right triangle?



yes

Problem 3

Find the missing leg (a) of the right triangle shown.



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